BrainStack: Heterogeneous Neural Ensembles for EEG-to-Language Decoding

Anonymous submission

Abstract

Electroencephalography (EEG) signals arise from the complex, nonlinear interplay of billions of neurons, posing significant challenges for reliable decoding. Despite advances in attention-based models, current methods often rely on global representations, limiting their ability to capture region-specific dynamics and inter-regional interactions. To address this, we propose BrainStack, a functionally guided heterogeneous ensemble framework for EEG-based text decoding. Motivated by neuroscientific findings, BrainStack partitions the brain into anatomically defined functional regions and jointly optimizes a global model alongside multiple region-specific models in an end-to-end fusion. The global branch, CTNet, employs a hybrid convolution–attention architecture to capture long-range dependencies and high-level contextual features, while the regional branches, CNets, use lightweight convolutional encoders to model localized activity. To effectively unify these heterogeneous perspectives, each upstream feature branch is fused through a gated meta-learner. We also introduce SilentSpeech-EEG (SS-EEG), a new large-scale benchmark comprising over 120 hours of EEG recordings from 12 subjects across 24 words, making it one of the largest datasets for silent speech decoding. BrainStack is evaluated under within-subject and cross-subject protocols, as well as on the public Thinking Out Loud dataset, confirming its robustness and generalization.

Code — https://btrrjl24.github.io/BrainStack/

Introduction

The human brain is a highly organized and distributed system composed of approximately 86 billion neurons. These neurons interact through complex electrical activity, and their collective dynamics give rise to macroscopic field potentials known as brainwave signals. Electroencephalography (EEG) provides a non-invasive and high-temporal-resolution technique for capturing large-scale cortical dynamics, making it particularly well-suited for real-time neural decoding (He 2013). With millisecond-level precision, modern EEG acquisition systems enable scalable and practical recording of neural signals across distributed brain regions (Lou et al. 2024). These capabilities form the foundation for brain-computer interface (BCI) applications, especially in cognitive and communication tasks where timely and accurate decoding of brain activity is essential (Liang et al. 2024a).

Despite its potential, decoding meaningful information from EEG remains fundamentally challenging due to the low signal-to-noise ratio, temporal non-stationarity, and the inherent heterogeneity of neural activity across individuals and brain regions (Choi 2025; Duan et al. 2023a).

Recent advances in deep learning have significantly improved the modeling capacity of EEG decoders. Convolutional and temporal models, such as CNNs, RNNs, and Transformers—have shown great potential in capturing complex spatiotemporal dependencies in EEG recordings (Wang et al. 2024b; Shi et al. 2023; Ding et al. 2025; Duan et al. 2023b; Cao et al. 2022). Moreover, trends like large-scale pretraining and self-supervised learning, which revolutionized natural language and vision domains, are starting to demonstrate encouraging results in neural decoding(Jiang et al. 2024; Wang et al. 2024a; Zhu et al. 2023; Tang et al. 2024; Svanera et al. 2021; Wu et al. 2022). Nevertheless, most existing EEG decoding models treat the brain as a homogeneous signal source, often ignoring the brain's intrinsic functional specificity. As a result, such models either overfit to the noisy global input or fail to leverage the complementary signals encoded in functionally distinct cortical regions (Liang et al. 2024b; Chen, Jia, and Wu 2024; Li and Fan 2020; Deng et al. 2023; Hao and Sun 2022). To mitigate this, recent work explores strategies such as multi-band modeling, region-specific attention, subject-adaptive learning, and dynamic neural representations (Shevchenko, Yeremeieva, and Laschowski 2024; Bian et al. 2024; Lou et al. 2024; Duggento et al. 2022).

The latest methods have adopted channel-wise tokenization strategies—treating each electrode as an independent sequence unit fed into Transformer backbones (Lee and Lee 2021; Ding et al. 2025). While this approach aligns well with the architectural design of attention mechanisms and enables efficient modeling of temporal dependencies, it lacks explicit modeling of anatomical clustering or spatial continuity. As a result, it overlooks the physiologically grounded inter-regional dependencies that shape neural representations (Shi et al. 2023; Choi 2025; Zheng et al. 2024). This becomes especially limiting in tasks like silent speech decoding, where subtle, distributed activation patterns across regions are essential for accurate classification (Wang et al.

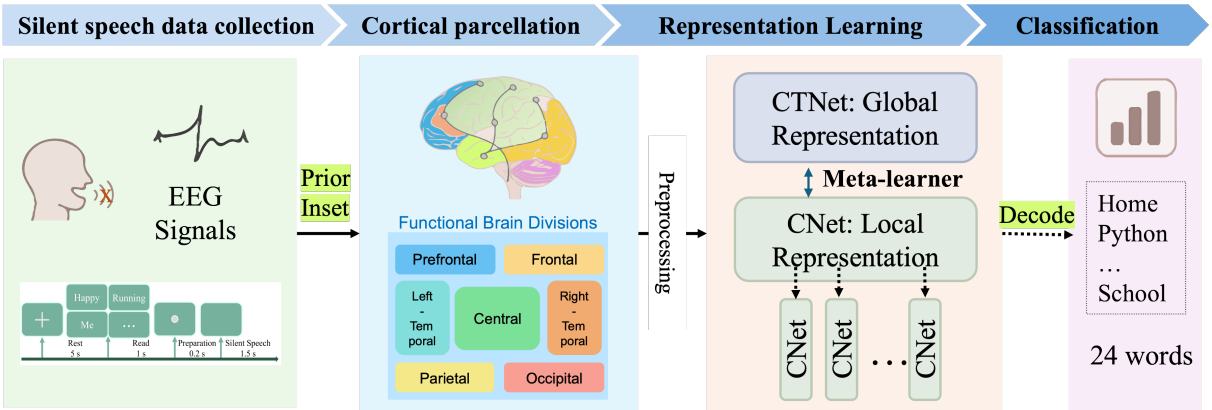


Figure 1: Overall illustration of EEG-based text decoding using BrainStack.

2020; Rekrut et al. 2021; Song et al. 2023; Fitriah, Zakaria, and Rajab 2022).

In contrast, neuroscience offers strong motivation to respect the brain's functional heterogeneity. Neuroscientific research has long emphasized the modular organization of the brain, with distinct cortical areas associated with specific cognitive roles (Lashley 2006), especially language (Luo et al. 2023; Jamali et al. 2024; Liu et al. 2023). Classical neuroanatomical studies consistently partition the cortex into regions that are functionally specialized, such as the prefrontal, temporal, and occipital lobes (Stephan et al. 2006). Neural responses during cognitive and linguistic tasks are inherently spatiotemporal and distributed, with different regions exhibiting task-specific activation patterns (Jiang et al. 2025; Anumanchipalli, Chartier, and Chang 2019; Cao et al. 2021). For example, the frontal lobe is often involved in executive control and speech motor planning, while temporal areas contribute to auditory and semantic processing (Li et al. 2022). These biological insights suggest that a functionally informed decoding framework could better exploit the spatial diversity of brain activity and lead to improved generalization (Johnston and Fusi 2024). While some recent studies have begun to incorporate region-aware modeling into EEG decoding, many still rely on simplified channel selection schemes or majority-voting ensembles, which are insufficient to capture the rich, cooperative dynamics across regions (Abbasi et al. 2020; Ding et al. 2022). Simple decision-level fusion strategies also struggle to generalize in low-data or high-noise regimes, where modeling the interaction between global and local representations becomes particularly important (Yue et al. 2017a). Graph-based approaches such as LGGNet have made initial progress toward incorporating region priors, but they generally adopt homogeneous encoders and static connectivity patterns, limiting their ability to capture dynamic coordination between scales (Ding et al. 2023).

To overcome these limitations, we introduce BrainStack, a functionally guided meta-ensemble learning framework designed for robust EEG-based neural decoding. Inspired by the modular organization of the human brain, BrainStack divides the EEG input into anatomically defined regions and simultaneously trains a global model alongside multiple region-specific models. The global branch captures long-range dependencies and contextual interactions across the whole brain using a hybrid convolution-attention architecture. In parallel, each regional branch focuses on modeling localized neural dynamics using lightweight convolutional encoders. To unify these heterogeneous perspectives, we introduce a Gated Expert Fusion method, where each upstream feature branch is treated as a task-specific expert. A learnable gating mechanism assigns adaptive weights to each expert, dynamically fusing their outputs before final prediction via a meta-learner MLP. This design enables task-aware expert selection and feature integration in an end-to-end manner.

Moreover, we introduce a knowledge distillation strategy where the global model serves as a teacher to the regional branches. By aligning their output distributions, this distillation mechanism encourages each regional model to learn from both local features and global semantic cues. This design not only enhances generalization under noisy or non-stationary conditions but also fosters cooperative learning between branches within the ensemble.

Our overarching objective is to develop an efficient and generalizable EEG decoder capable of leveraging complementary features across multiple brain regions. In doing so, BrainStack addresses the limitations of prior approaches by improving classification performance, reducing redundancy, and enhancing interpretability and deployment feasibility in practical brain-computer interface (BCI) applications. To summarize, our contributions are as follows.

- We propose BrainStack, a neuroscientifically guided ensemble framework that integrates global and region-specific EEG representations via a heterogeneous architecture and gated fusion.
- We introduce a multi-branch knowledge distillation scheme where the global model supervises regional branches, promoting cross-region consistency and robustness under subject variability.
- We present SS-EEG, a large-scale dataset for silent speech decoding with over 120 hours of EEG from 12 participants across 24 English words, providing one of

the most comprehensive benchmarks for word-level neural decoding.

Methodology

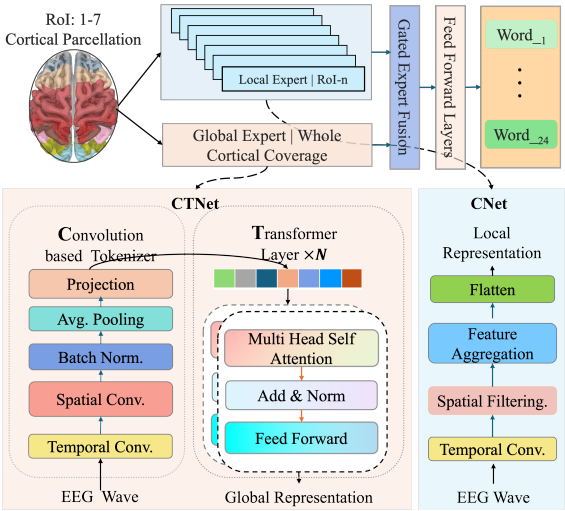


Figure 2: Overview of the proposed BrainStack for EEG-based text decoding. Raw signals $\mathbf{X} \in \mathbb{R}^{C\times T}$ are partitioned into seven region-specific subsets and one global input. Regional branches adopt lightweight spatio-temporal CNNs, while the global branch (CTNet) combines CNN and Transformer layers. Outputs are adaptively fused by a meta-learner f_{meta} for final prediction $y \in \mathcal{Y}$.

Problem Formulation

We formulate EEG-based neural decoding as a structured multi-class classification task. Each EEG trial is represented as a matrix $\mathbf{X} \in \mathbb{R}^{C\times T}$, where C denotes the number of channels and T is the number of time steps. The objective is to identify the corresponding cognitive or behavioral class label $y \in \{1, \ldots, K\}$. Unlike images or speech, EEG signals are inherently structured by anatomical priors. Channels can be grouped into N functionally defined brain regions, each associated with specific cognitive subfunctions and distinct signal characteristics. Let $\mathbf{X}_i$ denote the input segment corresponding to region i, and let its task-relevant latent representation $\mathcal{I}^i_{\text{local}}$ be defined as:

$$\mathcal{I}^i_{\text{local}} = \phi_i(\mathbf{X}_i), \tag{1}$$

where $\phi_i(\cdot)$ is a region-specific encoder. Due to anatomical and functional specialization, the regional representations are partially independent in their distribution and semantics.

However, cognitive processes rely on coordinated neural activity across anatomically distinct brain regions, where inter-regional complementarity is essential for capturing task-specific semantics. Since individual regions alone cannot fully represent the complexity of cognitive states, it is crucial to integrate distributed neural features into a unified, task-relevant representation. To account for task-specific variability—such as the prominence of temporal regions in language decoding and central motor regions in motor imagery—we model the global neural representation as a weighted aggregation of region-specific features, formally defined as:

$$\mathcal{I}_y = \sum_{i=1}^{N} \omega_i^y \cdot \mathcal{I}^i_{\text{local}} = \sum_{i=1}^{N} \omega_i^y \cdot \phi_i(\mathbf{X}_i), \tag{2}$$

where $\mathcal{I}_y$ represents the *task-specific global representation* that integrates region-specific features according to their contextual relevance under task y. The term ω_i^y denotes the learnable importance weight assigned to region i, reflecting its contribution to the task. These weights are dynamically optimized to emphasize anatomically distinct brain regions based on their neural significance.

Finally, the key challenge of EEG decoding lies in balancing regional specialization with cross-region integration, while adapting to task-specific variability. Formally, the decoding objective is to simultaneously learn both the local encoders $\phi_i(\cdot)$ and a fusion function $g(\cdot)$ to yield:

$$\hat{y} = \arg\max_k \left[g\left(\sum_{i=1}^{N} \omega_i^y \cdot \phi_i(\mathbf{X}_i) \right) \right]_k, \tag{3}$$

where k indexes the set of class labels, and $g(\cdot)$ maps the integrated features into class logits. The $\arg\max$ operator selects the class with the highest predicted score, resulting in the final output.

The Proposed Framework: BrainStack

Our framework builds on well-established neuroscientific evidence of functional brain specialization and is designed to provide a hierarchically structured representation pipeline that captures the multi-scale dynamics of neural activity.

Motivated by the hypothesis that cognitive functions are supported by both distributed and localized neural processes (Yue et al. 2017a), BrainStack explicitly decomposes the EEG input into one *global view* and seven *region-specific views* based on standard anatomical divisions: Prefrontal, Frontal, Central, Left-Temporal, Right-Temporal, Parietal, and Occipital (Liang et al. 2024b). This decomposition is not arbitrary—it provides an inductive bias for learning specialized regional features while preserving access to whole-brain interactions, a design that has been shown to enhance decoding performance in EEG-based BCIs (Lou et al. 2024). Each EEG trial $\mathbf{X} \in \mathbb{R}^{C\times T}$ is processed in parallel by a global encoder and seven local encoders, each receiving a tailored spatial subset $\mathbf{X}_i \in \mathbb{R}^{C_i\times T}$.

Heterogeneous Model Design To accommodate the distinct signal characteristics and modeling goals of global versus regional inputs, BrainStack adopts a heterogeneous encoder structure—a carefully designed architectural aimed at enhancing both representational power and computational efficiency in multi-scale EEG modeling (Zhang et al. 2023; Ma, Song, and Gao 2022).

CTNet: Attention-based Global Encoder As illustrated in Fig. 2, the global branch is designed to capture long-range dependencies across the full electrode montage by combining convolutional feature extraction with Transformer-based sequence modeling. The input is a preprocessed EEG trial $\mathbf{X} \in \mathbb{R}^{C \times T}$, where C denotes the number of channels and T the number of time steps. The encoder operates in two main stages: *patch embedding* and *Transformer encoding*.

1) Patch Embedding. We implement a convolutional embedding module that transforms continuous EEG signals into a sequence of latent tokens. A temporal convolution with kernel size $(1, L)$ extracts short-duration dynamics within each channel, followed by a spatial convolution with kernel size $(C, 1)$ that aggregates information across all electrodes, effectively learning inter-channel dependencies. Batch normalization and an ELU activation improve stability, and average pooling along the temporal axis segments the signal into fixed-length windows. Each window corresponds to a *multi-channel temporal patch*, rather than a single-channel slice. A final 1×1 convolution projects these patches into a D-dimensional embedding, yielding a token sequence $\mathbf{Z} \in \mathbb{R}^{N \times D}$, where N is the number of patches.

2) Transformer Encoder. The patch tokens are then processed by a stack of Transformer layers, each consisting of multi-head self-attention and feedforward blocks with residual Add–Norm connections. This stage integrates information across the entire sequence, enabling the model to capture long-range temporal dependencies and cross-channel correlations beyond the receptive field of convolutional filters.

The global encoder therefore outputs a set of contextualized representations that summarize spatiotemporal dynamics across the whole brain. These representations are subsequently forwarded to the gated fusion module, where they are combined with regional encoders for final prediction.

CNet: Convolution-based Local Encoders To complement the global encoder, BrainStack incorporates a series of lightweight region-specific branches, each designed to process EEG signals from a distinct functional area of the brain. For each region $n \in \{1, ..., N\}$ (where $N = 7$), the corresponding input $\mathbf{X}_n \in \mathbb{N}^{C_n \times T}$ is selected via a predefined anatomical mapping, ensuring that $C_n \ll C$ and that each encoder focuses on localized neural dynamics.

Inspired by *EEGNet* architecture (Lawhern et al. 2018), each local encoder adopts a compact convolutional design tailored for EEG decoding. It consists of temporal convolutions to capture short-duration patterns, depthwise spatial convolutions to extract inter-channel correlations within a region, and separable convolutions for efficient feature abstraction. Batch normalization and dropout are applied throughout to ensure stable convergence and mitigate overfitting.

This design encourages each local encoder to specialize in learning discriminative features within its anatomical context while allowing for scalable parallel computation across regions. All encoders, both global and local, are trained jointly in an end-to-end manner to optimize coordinated feature learning and classification (Chen, Jia, and Wu 2024).

Gated Expert Fusion with Meta Learner Instead of naive averaging, BrainStack introduces a Meta Learner to adaptively integrate features across the eight branches (global + seven local):

$$F_{\text{meta}} = \sum_{i=1}^{N} \alpha_i \cdot F_i, \tag{4}$$

where F_i represents the learned representations from each upstream expert, and α_i is a trainable fusion weight optimized end-to-end and constrained via softmax normalization, $\sum_{i=1}^{N} \alpha_i = 1$. This design allows the model to dynamically emphasize different spatial and temporal components during feature integration, enabling richer contextual awareness and improved robustness against noise or artifacts in EEG signals.

To achieve this, each encoder's feature representation is adaptively weighted according to its contextual relevance, determined through a lightweight attention mechanism:

$$\alpha_i = \frac{\exp(h(F_i))}{\sum_{j=1}^{N} \exp(h(F_j))}, \tag{5}$$

where $h(\cdot)$ is a learned projection that assesses the contribution of each feature map to the final decision. The fused meta-representation F_{meta} is subsequently projected into the logit space through a fully connected layer before final softmax activation.

Dynamic Multi-Objective Loss Function

We adopt a staged and dynamically scheduled strategy. Training begins with a global warm-up phase and gradually transitions to full supervision, including fused prediction, global guidance, regional specialization, and knowledge distillation (Du et al. 2020; Zoumpourlis and Patras 2024). The overall objective is to minimize a hybrid loss that combines four components: $\mathcal{L}_{\text{fused}}$ for the fused prediction, $\mathcal{L}_{\text{global}}$ for the global model, $\mathcal{L}_{\text{local}}$ for all local experts, and $\mathcal{L}_{\text{distill}}$ for knowledge distillation from the fused output to the local models (Tu, Liu, and Xiao 2022).

Formally, the overall optimization objective is:

$$\mathcal{L}_{\text{total}} = \lambda \cdot \mathcal{L}_{\text{fused}} + \alpha \cdot \mathcal{L}_{\text{global}} + \beta \cdot \mathcal{L}_{\text{local}} + \gamma \cdot \mathcal{L}_{\text{distill}}, \tag{6}$$

where λ, α, β, and γ are hyperparameters controlling the relative weights of each term.

During warmup, we begin by prioritizing the global branch to establish a stable embedding. Loss weights are then scheduled dynamically based on training progress, defined as:

$$\mathcal{P}(t) = \min\left(1.0, \frac{\text{epoch} - T_{\text{warmup}}}{T_{\text{transition}}}\right), \tag{7}$$

where $\mathcal{P}(t)$ denotes the training progress at epoch t, T_{warmup} represents the warm-up phase duration, and $T_{\text{transition}}$ indicates the length of the transition period.

The weight of the fused loss is then adaptively scheduled as:

$$\lambda_{\text{fused}} = (1 - \mathcal{P}(t)) \cdot \lambda_{\text{min}} + \mathcal{P}(t) \cdot \lambda_{\text{max}}, \tag{8}$$

where λ_{min} and λ_{max} define the lower and upper bounds for the fusion weight.

Table 1: Comparison of SS-EEG with representative EEG datasets for imagined or silent speech decoding.

Dataset	Subjects	Vocabulary / Classes	Trials	Duration	Channels	Sessions	Repeats per Class
KaraOne (Zhao and Rudzicz 2015)	12 (8 usable)	11 (7 phonemes + 4 words)	1056	∼4.5 h	64 EEG + 4 EOG	3	12
Thinking Out Loud (Nieto et al. 2022)	10	4 (Spanish words)	2236	∼3.5 h	128 EEG + 8 EXG†	3	55–60
SS-EEG (Ours)	12 (10 usable)	24 (English words)	60,000	∼120 h	128 EEG + 8 EXG†	16	250

EXG channels include 4 auxiliary electrodes for eye (EOG) and 2 muscle (EMG) activity monitoring, as well as 2 reference electrodes.

For auxiliary supervision, we define a shared scheduling function:

$$\omega_{\text{aux}}(x) = x \cdot \mathcal{P}(t) \cdot \left(1 - \min\left(\frac{\mathcal{L}_{\text{fused}}}{\text{max_loss_estimate}}, 1.0\right)\right) \quad (9)$$

where $\omega_{\text{aux}}(x)$ denotes the weight assigned to auxiliary component x, $\mathcal{P}(t)$ is the training progress, $\mathcal{L}_{\text{fused}}$ is the current fused loss, and max_loss_estimate is an empirical upper bound on the loss.

Thus, the final loss weights become:

$$\mathbf{w} = \left[\omega_{\text{aux}}(\alpha_{\max}),\, \omega_{\text{aux}}(\beta_{\max}),\, \omega_{\text{aux}}(\gamma_{\max}) \cdot \mathcal{P}(t)\right], \quad (10)$$

where $\mathcal{P}(t)$ scales the importance of these weights dynamically during training.

We adopt a knowledge distillation loss to propagate high-level semantic knowledge from the global expert to regional experts:

$$\mathcal{L}_{\text{distill}} = \sum_{i=1}^{N} \text{KL}\left(\text{Softmax}\left(\frac{\mathbf{F}_{\text{global}}}{T}\right) \parallel \text{Softmax}\left(\frac{\mathbf{F}_i}{T}\right)\right) \quad (11)$$

where $\mathbf{F}_{\text{global}}$ and $\mathbf{F}_i$ denote the logits from the global model and the i-th regional model, respectively. This objective transfers high-level predictive signals from the global branch to each local expert, enhancing their capacity to model nuanced patterns while maintaining coherence with the fused prediction.

Experiments

Dataset

Silent Speech EEG (SS-EEG) To evaluate our proposed framework, we collect and introduce SS-EEG, a new large-scale dataset designed for silent speech decoding from EEG signals. The dataset comprises over 120 hours of EEG recordings from 12 participants, each performing covert articulation of 24 commonly used English words (e.g., "go", "happy", "python", "running"). Designed for both classification-based decoding and future language model pretraining, SS-EEG provides a rich, large-scale foundation for advancing EEG-to-text research.

Each subject completed 16 sessions, totaling 6,000 trials per participant. A single session contains 375 silent speech trials and lasts approximately 40 minutes including scheduled breaks. The experimental paradigm follows a four-stage protocol: *rest*, *read*, *preparation*, and *silent speech*. Each trial begins with a 5-second rest phase, including a 1.5-second fixation cross. The target word is then displayed for 1 second during the read phase. A brief 0.2-second auditory cue signals transition into the imagination phase, where the subject silently repeats the word for 1.5 seconds. To minimize fatigue, participants are provided with a break after every 20 trials for up to 5 minutes. EEG signals were recorded using a 128-channel Neuroscan system at a 1000 Hz sampling rate. Signals were band-pass filtered to 1–45 Hz and segmented into clean trial epochs. We further applied per-channel z-score normalization and removed trials with excessive noise or artifacts. It is noted that due to severe signal contamination in two subjects, the final released dataset includes recordings from 10 participants (subject-01 to subject-10).

To the best of our knowledge, SS-EEG is the largest publicly available dataset for word-level silent speech decoding using EEG, and we will release it upon publication.

Baseline and Metrics

To avoid temporal leakage, we adopt a session-wise evaluation protocol. For each subject, the 16 EEG sessions are split into 14 for training, 1 for validation, and 1 for testing. This ensures that training and test data come from non-overlapping temporal blocks, thereby preventing adjacent trials from leaking temporal information across splits.

For benchmarking, we compare BrainStack against a representative suite of baselines, including two fully supervised CNN-based models EEGNet (Lawhern et al. 2018) and TCNet (Ingolfsson et al. 2020), two Transformer-based models EEGConformer (Song et al. 2022) and STTransformer (Song et al. 2021), and a pre-trained model, LaBraM (Jiang, Zhao, and Lu 2024). All baselines are trained under identical settings within the same experimental framework for fair comparison.

Beyond the within-subject evaluation on SS-EEG, we also conduct supplementary experiments under a cross-subject protocol (leave-one-subject-out) and on the public *Thinking Out Loud* dataset (Nieto et al. 2022). These results are reported in the Appendix under Additional Experiments: Cross-Subject and Public Dataset, further confirm the robustness of our framework.

Performance Comparison

Table 2 summarizes the classification performance of BrainStack and a diverse set of baseline models on the SS-EEG

Table 2: Performance comparison on the SS-EEG dataset for word-level classification.

Model	Param.	S01	S02	S03	S04	S05	S06	S07	S08	S09	S10	Avg. Acc
EEGNet	**8.54K**	40.80%	35.30%	33.50%	22.90%	**17.70%**	10.60%	49.60%	14.00%	38.70%	24.70%	28.78%
TCNet	78.62K	36.90%	36.40%	32.70%	22.70%	14.30%	15.30%	44.40%	16.40%	50.40%	25.50%	29.50%
EEGConformer	0.75M	58.20%	32.70%	27.50%	12.20%	11.20%	7.00%	30.90%	7.00%	23.90%	28.30%	23.89%
STTransformer	2.78M	60.30%	16.40%	30.90%	21.30%	17.40%	**35.10%**	15.60%	**31.20%**	31.20%	23.40%	28.28%
LaBraM	5.80M	63.40%	13.00%	15.10%	9.10%	8.60%	7.50%	34.30%	8.30%	13.20%	10.40%	18.29%
BrainStack_Homo	5.93M	81.04%	27.79%	32.98%	23.37%	6.23%	11.69%	53.77%	11.69%	60.77%	18.44%	32.78%
BrainStack_RoI5	1.05M	80.26%	39.22%	44.15%	24.94%	5.71%	11.42%	56.10%	15.84%	62.85%	31.42%	37.19%
BrainStack	1.06M	**88.05%**	**39.48%**	**48.57%**	**29.87%**	10.13%	17.40%	**60.25%**	19.74%	**70.64%**	**34.54%**	**41.87%**

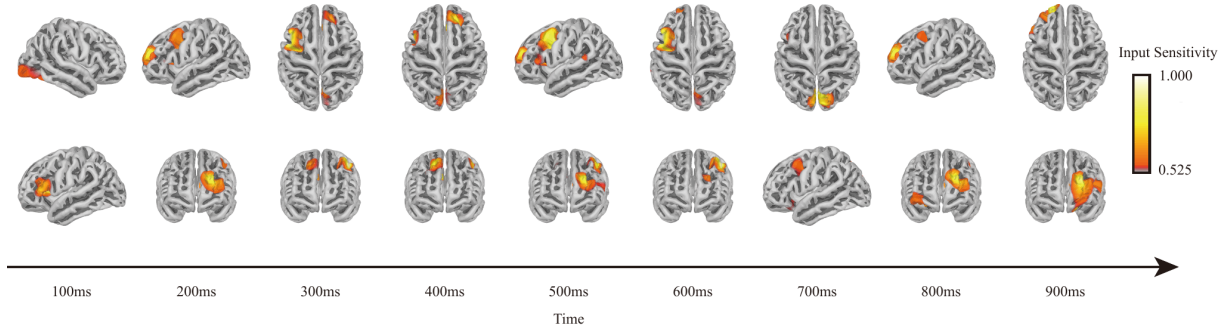


Figure 3: **Human attention dynamics during silent speech execution.** Temporal evolution of brain activation maps obtained through attribution analysis. The maps illustrate the spatiotemporal distribution of sensitivity in response to the input stimulus across different brain regions from 100ms to 900ms. Warmer colors represent higher sensitivity, indicating stronger contribution to the model's decision at each time step. Each vertical pair of cortical maps represents two complementary views of the brain at the same moment, offering a clearer visualization of region-specific activations. The analysis captures the progression of localized neural responses transitioning into broader inter-regional integration, revealing both independent regional dynamics and global coordination underlying silent speech decoding.

dataset. Among the baselines, EEGNet and TCNet represent compact CNN-based architectures, while EEGConformer and STTransformer are attention-based designs. LaBraM serves as a large-scale pretrained foundation model.

BrainStack achieves the best overall performance with an average accuracy of 41.87%, outperforming all baselines and model variants. Compared to the strongest baseline (TCNet, 29.50%), BrainStack delivers a significant improvement of +12.37%. Notably, it achieves the highest accuracy on 8 out of 10 subjects, and remains highly competitive on the remaining ones, reflecting strong generalization across individuals.

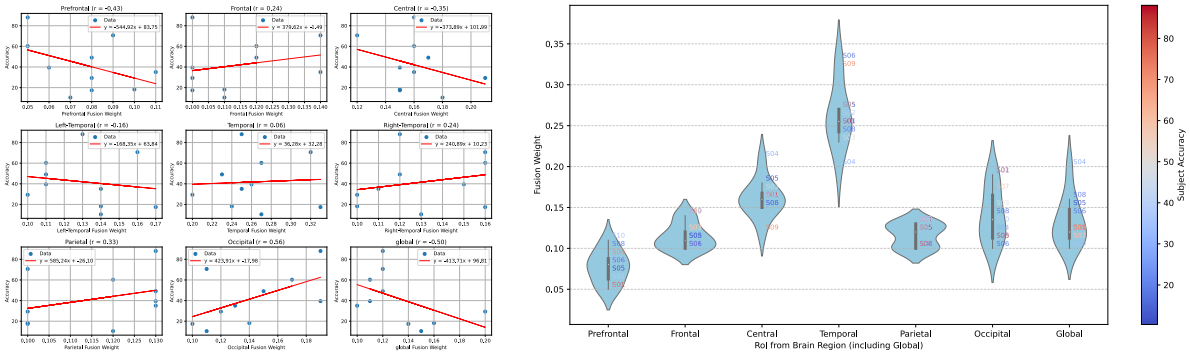
Compared to BrainStack_ROI5 (37.19%), which adopts a coarse 5-region anatomical partition, BrainStack achieves a +4.68% improvement, demonstrating the benefit of fine-grained, functionally informed cortical decomposition. Meanwhile, BrainStack_Homo, which uses identical CTNet encoders across all branches, performs worse (32.78%) despite having over five times the parameter count. These comparisons highlight that structural diversity and functional alignment are more crucial than model scale for capturing spatially distributed EEG patterns. Notably, BrainStack surpasses LaBraM (18.29%) by a wide margin, while using only one-fifth of the parameters (1.06M vs. 5.80M), underscoring the importance of task-specific architectural design over model size alone in low-SNR neural decoding.

To understand the contribution of each brain region to decoding performance, we analyze the learned fusion weights. Fig. 4(a) shows that Occipital fusion weights are strongly correlated with subject-level accuracy ($r = 0.56$), highlighting the role of visual cortical signals. In contrast, the Global branch shows a moderate negative correlation ($r = -0.50$), suggesting that over-reliance on holistic features may hinder performance. Prefrontal and Central regions also exhibit negative trends, while Parietal and Frontal regions are weakly positive. Temporal fusion weights, though showing minimal correlation ($r = 0.06$), are consistently high across subjects.

As seen in Fig. 4(b), high-performing subjects (e.g., S05, S09) tend to assign more weight to Temporal and Occipital regions, indicating their combined predictive and complementary roles. These findings suggest that our meta-learner not only favors predictive regions but also leverages stable contributors like Temporal for generalization.

Ablation Study

To evaluate the contributions of individual components in our framework, we conduct ablation studies on three key aspects of BrainStack: (1) the ensemble architecture, (2) the fusion mechanism, and (3) the loss design. Results across



(a) (b)

Figure 4: Regional contribution analysis in the gated fusion module. **(a)** Correlation between average fusion weight of each region and subject-level accuracy, with fitted regression lines and Pearson's r. **(b)** Violin plots of fusion weight distributions across regions, overlaid with subject accuracies (color-coded), highlighting variability in regional importance.

Table 3: Ablation study on ensemble architecture, fusion mechanism, and loss design.

Objective	Model	Param.	S01	S02	S03	S04	S05	S06	S07	S08	S09	S10	Avg. Acc
Ensemble architecture	Local only-CNet	**0.18M**	78.44%	28.83%	22.59%	17.14%	6.23%	4.41%	53.77%	12.47%	57.14%	27.53%	30.86%
	Homogeneous-CNet	0.20M	70.65%	34.02%	36.36%	22.86%	4.94%	3.38%	55.06%	12.47%	62.08%	34.29%	33.61%
	Global only-CTNet	1.01M	72.47%	28.05%	44.41%	24.94%	8.31%	10.38%	56.62%	15.32%	65.71%	28.83%	35.50%
	Homogeneous-CTNet	5.93M	81.04%	27.79%	32.98%	26.75%	6.23%	11.69%	53.77%	11.69%	60.77%	18.44%	33.12%
Fusion Mechanism and Loss Design	Token-level fusion	5.21M	82.33%	29.61%	32.98%	14.28%	9.09%	11.17%	34.55%	12.73%	42.86%	24.45%	29.41%
	w/o knowledge distillation	1.06M	84.94%	29.87%	38.70%	17.14%	5.97%	14.03%	54.03%	14.55%	70.38%	27.27%	35.69%
	w/o warm-up	1.06M	84.42%	34.54%	29.87%	12.98%	7.01%	12.47%	56.62%	16.6%	55.58%	29.61%	33.97%
Ours	**BrainStack**	1.06M	**88.05%**	**39.48%**	**48.57%**	**29.87%**	**10.13%**	**17.40%**	**60.25%**	**19.74%**	**70.64%**	**34.54%**	**41.87%**

ten subjects are reported in Table 3.

Ensemble architecture. We compare four configurations to assess the role of model structure. The *Local only-CNet* setup aggregates outputs from independent region-specific CNNs and achieves 30.86%, confirming the limited capacity of purely local modeling. Adding structural regularity via *Homogeneous-CNet* improves performance slightly (33.61%), while *Global only-CTNet* achieves a stronger 35.50%, validating the value of holistic context modeling. The best variant among these is *Homogeneous-CTNet* (33.12%), which uses the same architecture across all branches. However, none of these match the performance of the full BrainStack, which integrates both global and regional perspectives through a heterogeneous design, reaching 41.87%. These comparisons demonstrate that architectural diversity and functional partitioning are more effective than either approach in isolation.

Fusion mechanism and loss design. To evaluate the impact of our adaptive fusion and training objective, we test token-level fusion and partial loss configurations. The token-level fusion strategy in BrainStack adopts an attention-based meta-learner that aggregates intermediate token representations across all branches, enabling fine-grained integration of spatially distributed features. However, when applied in isolation without heterogeneous modeling or sufficient contextual depth, this fusion mechanism performs worse (29.41%), likely due to limited expressiveness of shallow representations. Removing knowledge distillation or the warm-up schedule also leads to performance drops (35.69% and 33.97%, respectively), confirming that global-to-local regularization and progressive optimization are critical for stable convergence and effective multi-branch coordination.

Conclusion

In this paper, we presented BrainStack, a functionally guided ensemble framework for eeg based nerual language decoding, which integrates global and regional neural representations through heterogeneous encoders and gated meta-learner. To stabilize training, we introduced a dynamic multi-objective loss with progressive scheduling, enabling effective coordination of global supervision, regional specialization, and knowledge distillation. We further released SilentSpeech-EEG (SS-EEG), a new benchmark with over 120 hours of EEG recordings, making it the largest dataset for silent speech decoding to date. Extensive evaluations under within-subject, cross-subject, and on the Thinking Out Loud dataset demonstrate that BrainStack achieves state-of-the-art performance and strong generalization, providing a scalable and interpretable foundation for advancing neural language decoding in BCI systems.

Appendices

Additional Subject-Dependent Results

A.1 Statistical significance

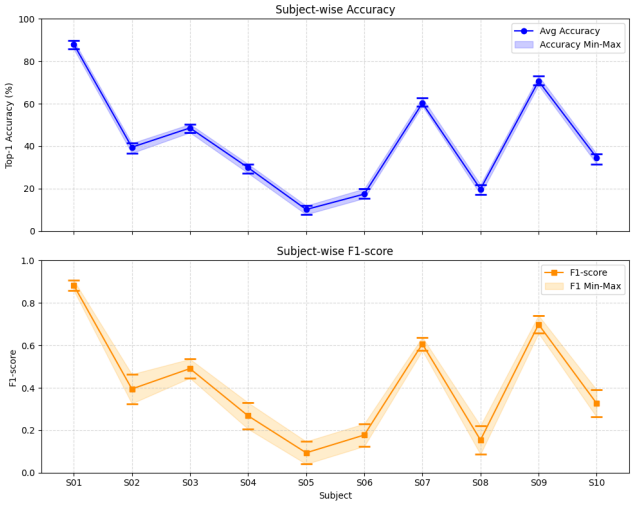


Figure 5: Subject-wise performance variability across three random seeds. Top: classification accuracy with min–max error bands. Bottom: F1-score with corresponding min–max range.

To evaluate subject-specific performance consistency, we report the classification accuracy and F1-score for each of the 10 test subjects, averaged over three independent random seeds. As visualized in Figure 5, the results reveal considerable variability in model performance across subjects.

High-performing subjects such as **S01**, **S07**, and **S09** achieved consistently strong decoding performance, with average accuracies above 60% and F1-scores exceeding 0.60. These subjects also exhibited narrow error margins, suggesting robust model behavior under different random initializations. In particular, S01 achieved the highest performance with an average accuracy of 88.05% and F1-score of 0.88.

In contrast, low-performing subjects including **S05**, **S06**, and **S08** showed both low average performance (e.g., S05: 10.13% accuracy, 0.09 F1) and substantial seed-dependent fluctuation. These results indicate sensitivity to trial-level noise and signal quality, emphasizing the challenge of decoding from weak or inconsistent neural responses.

The error bars in Figure 5 delineate a clear pattern: consistent performance among high-performing subjects and pronounced variability among lower-performing ones. Such disparity underscores the effectiveness of ensemble learning and spatially informed decomposition in enhancing model robustness and generalization across subject-specific neural diversity.

A.2 Spatial neural patterns

To further understand the spatial neural patterns underlying the observed subject-wise performance, we present a region-wise interpretability analysis of the BrainStack model using a clustered heatmap of normalized importance scores (Figure 6). Each row represents a subject and each column corresponds to a predefined brain region, with both dimensions hierarchically clustered to reveal latent structure.

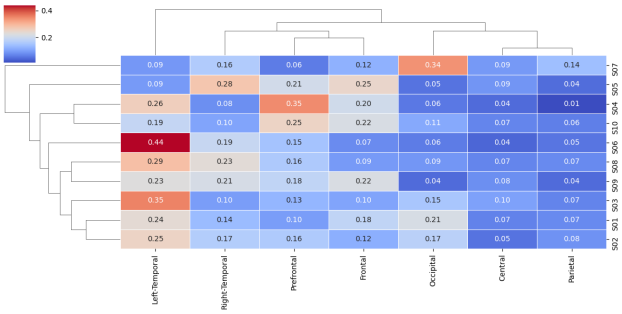


Figure 6: Clustered heatmap of normalized regional contributions across subjects. Both rows (subjects) and columns (brain regions) are hierarchically clustered.

The results show that the left and right temporal lobes consistently receive higher attribution scores across subjects, indicating their dominant role in silent speech decoding. This observation aligns with established neuroscientific evidence that the temporal cortex is essential for phonological processing and auditory representation, even in the absence of overt speech.

In contrast, parietal and central regions exhibit generally lower importance, suggesting limited involvement in this decoding task. Interestingly, subjects with higher attribution in temporal and prefrontal areas tend to cluster together, possibly reflecting shared neural strategies or more robust internal speech representations.

Moreover, the clustering patterns reveal subject-level heterogeneity in regional reliance, highlighting the importance of individualized neural decoding strategies. These findings provide neurophysiologically plausible explanations for model behavior and further motivate region-aware or subject-adaptive modeling approaches in silent speech BCI systems.

A.3 Word-level confusion analysis

Figure 7 shows the confusion matrices for all 10 subjects across the 24 word-level classes. High-performing subjects such as S01, S07, and S09 exhibit clear diagonal dominance, indicating accurate and confident predictions. The sparse off-diagonal elements suggest that errors, when present, are isolated rather than systematic.

In contrast, low-performing subjects like S05 and S06 display highly dispersed confusion patterns, with predictions distributed across many incorrect labels. This indicates a lack of discriminative signal or high trial-level variability, and aligns with the low accuracy and F1-scores reported in earlier sections.

Interestingly, some subjects (e.g., S03, S10) show moderate performance with partial diagonal patterns and local confusion clusters, hinting at class-dependent decoding difficulty. This could reflect the neural similarity of certain words or semantic categories, such as motion-related or number-related terms. Future work may benefit from semantic-aware

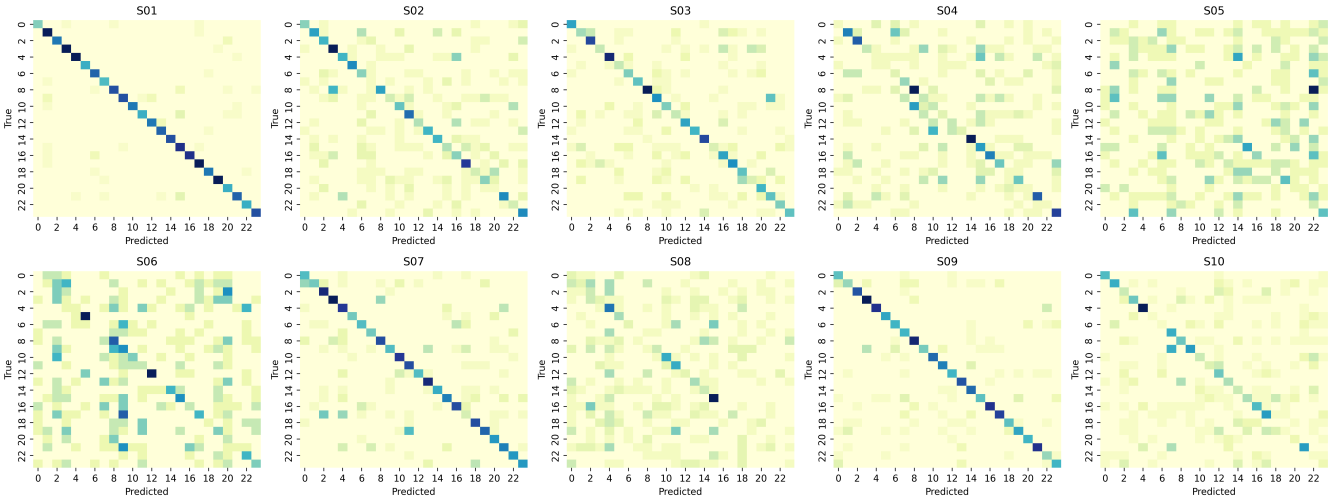


Figure 7: Confusion matrices for all 10 subjects in the SS-EEG dataset. Rows indicate true labels and columns indicate predicted labels (24 words total).

Table 4: Performance comparison on the SS-EEG dataset under the cross-subject setting (leave-one-subject-out).

Model	**S01**	**S02**	**S03**	**S04**	**S05**	**S06**	**S07**	**S08**	**S09**	**S10**	**Avg. Acc.**
EEGNet	10.83%	7.52%	5.05%	5.00%	4.78%	5.20%	4.52%	4.82%	5.75%	5.30%	5.88%
TCNet	5.97%	6.50%	6.07%	4.89%	5.62%	7.13%	6.25%	4.55%	7.35%	7.18%	6.15%
EEGConformer	6.91%	7.55%	8.27%	5.10%	5.53%	7.11%	10.28%	4.73%	9.05%	7.33%	7.19%
BrainStack	**13.42%**	**8.27%**	**11.25%**	**5.13%**	**6.60%**	**8.05%**	**11.97%**	**5.10%**	**10.18%**	**8.41%**	**8.84%**

decoding architectures or adaptive label grouping strategies.

Additional Experiments: Cross-Subject and Public Dataset

While the main paper focuses on within-subject evaluation with SS-EEG, we also conducted supplementary experiments to assess generalization across subjects and datasets. Specifically, we include (i) cross-subject decoding under a leave-one-subject-out (LOSO) protocol on SS-EEG, and (ii) evaluation on the public *Thinking Out Loud* dataset. Here we report the overall comparative results; more detailed analyses (e.g., ablations, regional contributions, error breakdowns) are restricted to the within-subject SS-EEG setting.

Cross-subject evaluation. Table 4 reports the results under a leave-one-subject-out (LOSO) protocol on the SS-EEG dataset. Compared to the within-subject evaluation, the overall accuracies drop substantially in the cross-subject setting, which is a well-known challenge in EEG decoding. This degradation mainly arises from the strong physiological variability and non-stationarity of EEG signals across individuals. Such differences are not limited to channel distributions and noise patterns, but also extend to individual-specific linguistic representations and cognitive habits, leading to a pronounced domain shift between training and test subjects.

Despite the low absolute accuracy, BrainStack consistently achieves the best overall performance with an average accuracy of 8.84%, outperforming CNN-based models (EEGNet: 5.88%, TCNet: 6.15%) and Transformer-based models (EEGConformer: 7.19%). These results indicate that even under the more challenging cross-subject protocol, the proposed heterogeneous ensemble with gated fusion can capture certain cross-participant invariant features and demonstrate relatively stable performance advantages.

While these findings highlight the inherent difficulty of cross-subject EEG-to-text decoding, they also point to potential directions for future research. We argue that further improvements hinge on reducing the distributional gap between individuals, which may be addressed by domain adaptation techniques or by explicitly modeling and aligning subject-specific characteristics. In addition, parameter-efficient transfer strategies, such as lightweight adapter modules, could enable fast adaptation to unseen participants while maintaining a shared backbone, and multimodal fusion may introduce external priors (e.g., EMG or speech signals) to provide additional constraints.

Thinking Out Loud dataset. Table 5 presents the results on the public *Thinking Out Loud* dataset under a within-subject evaluation protocol. Compared with our self-collected SS-EEG dataset, the overall accuracies on this cor-

Table 5: Performance comparison on the Thinking Out Loud dataset.

Model	S01	S02	S03	S04	S05	S06	S07	S08	S09	S10	Avg. Acc.
EEGNet	33.33%	30.56%	36.67%	**36.11** %	30.56%	33.33%	26.67%	40.00%	33.33%	30.56%	33.11%
TCNet	26.67%	**36.11%**	36.67%	27.78%	27.78%	27.27%	27.78%	26.67%	30.56%	33.33%	30.06%
EEGConformer	40.00%	27.78%	30.00%	30.56%	27.78%	30.30%	27.78%	36.67%	36.11%	33.33%	32.03%
BrainStack	**40.00%**	30.56%	**36.67%**	27.78%	**33.33%**	**40.00%**	**30.56%**	**40.00%**	**36.11%**	**33.33%**	**34.83%**

pus are lower, which can be attributed to its limited scale and task design. Specifically, the vocabulary size is restricted to four Spanish words, and the relatively small number of repetitions per class makes it difficult for models to learn stable discriminative patterns within each subject. The short recording duration further constrains the data available for training, increasing the risk of overfitting and noise sensitivity. In addition, cross-linguistic differences, such as the phonological structure of Spanish words, may also contribute to variability across subjects.

Under these constraints, BrainStack still achieves the highest average accuracy of 34.83%, slightly outperforming EEGNet (33.11%) and EEGConformer (32.03%), and clearly surpassing TCNet (30.06%). Although the absolute improvement is modest, the consistent advantage across subjects indicates that the proposed heterogeneous ensemble with gated fusion can effectively leverage complementary local and global features to maintain stable performance.

Taken together, the experiments on SS-EEG under a cross-subject protocol and the public Thinking Out Loud dataset provide strong evidence that BrainStack is not narrowly tailored to a single dataset or evaluation scheme. In the highly challenging cross-subject setting, where accuracies are generally low due to substantial inter-individual variability, BrainStack consistently outperforms representative CNN- and Transformer-based baselines, demonstrating its ability to capture partially transferable neural representations across participants. On the public dataset, despite its smaller vocabulary, limited recording duration, and linguistic differences from SS-EEG, BrainStack again achieves the best overall performance. These complementary results indicate that the proposed functionally guided ensemble framework generalizes beyond the SS-EEG corpus, alleviating concerns about dataset-specific adaptation and highlighting its potential as a robust foundation for broader EEG-to-text decoding applications.

The SS-EEG Dataset Details

The SilentSpeech-EEG (SS-EEG) dataset was developed to support non-invasive silent speech decoding research based on electroencephalography (EEG). The dataset comprises over 120 hours of EEG recordings from 12 right-handed native English speakers.

C.1 Participants

- **Total Subjects:** 12 right-handed native English speakers (6 male, 6 female)
- **Age:** Mean = 28.0 years, SD = 0.85 years
- **Health Status:** No reported visual, speech, or neurological disorders
- **Recruitment and Consent:** All participants were recruited in Australia and provided written informed consent
- **Ethics Approval:** Approved by the IRB ethics committee

C.2 Experimental environment and task design

- **Setup:** Participants sat in a quiet lab, facing a monitor at a comfortable distance
- **Instructions:** Subjects kept their eyes open and minimized body movement
- **Task:** Silent articulation (no sound or subvocalization) of prompted words
- **Vocabulary:** 24 words from 6 semantic categories: *motion*, *emotion*, *location*, *person*, *object*, *number*
- **Trial Design:** Each word repeated 5 times per session; randomized order
- **Implementation:** Task presented via PsychoPy (Python)

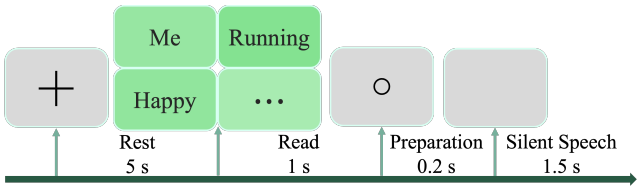


Figure 8: Schematic of a single trial in the silent speech EEG (SS-EEG) experiment. Each trial consists of four phases: a rest period (5 s) with fixation, word display (1 s), an auditory cue (0.2 s), and the silent articulation phase (1.5 s).

C.3 Experimental pipeline

- **Total Duration:** Over 120 hours of EEG data
- **Sessions:** Each subject completed 16 sessions, totaling 6,000 trials per participant
- **Session Length:** Each session includes 375 trials and lasts approximately 40 minutes (including breaks)
- **Trial Phases:** Each trial follows a four-stage protocol:
  - *Rest (5 s)* — includes a 1.5 s fixation cross
  - *Read (1 s)* — target word displayed on screen
  - *Preparation (0.2 s)* — auditory cue signaling onset

- *Silent Speech (1.5 s)* — subject imagines articulating the word

- **Fatigue Mitigation:** 5-minute break every 20 trials
- **Vocabulary:** 24 English words across 6 semantic groups (*e.g.,* “go”, “happy”, “python”, “running”)
- **Final Dataset:** Due to severe signal contamination, 2 subjects were excluded. Final release includes data from 10 subjects (Subject-01 to Subject-10)

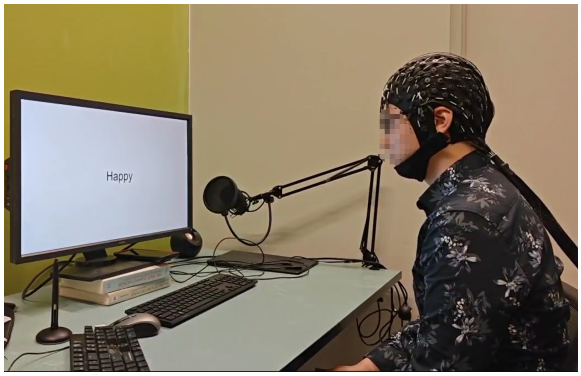


Figure 9: Experimental setup for silent speech EEG recording. A participant silently articulates the displayed word while wearing a 128-channel Neuroscan EEG cap. Recordings were performed in a controlled, sound-attenuated lab environment.

C.4 Recording setup

- **EEG Cap:** 128-channel Neuroscan Quik-Cap®
- **Amplifier:** SynAmps RT 128-channel amplifier[1]
- **Software:** CURRY 9.0 for acquisition[2]
- **Montage:** 133 channels total:
  - 122 EEG channels (brain activity)
  - 4 EOG electrodes (eye movement)
  - 6 reference electrodes (environment/muscle noise)
  - 1 trigger channel (condition labeling)
- **Electrode Placement:** 10–10 international system
- **Impedance:** Maintained below 10 kΩ and balanced across channels

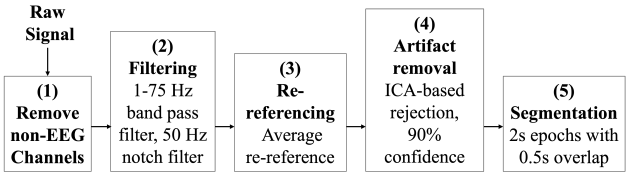


Figure 10: EEG preprocessing pipeline.

[1]https://compumedicsneuroscan.com/product/synamps-rt-128-channel-eeg-erp-epamplifier/

[2]https://compumedicsneuroscan.com/product/curry-9-x-data-acquisition-and-online-processing/

C.5 Preprocessing

- **Filtering:** 1–75 Hz bandpass (FIR), 50 Hz notch filter for line noise
- **Referencing:** Common average across 122 EEG channels
- **Artifact Removal:** ICA-based rejection of eye, muscle, and cardiac noise; 90% confidence threshold
- **Segmentation:** Epochs of 2 s length with 0.5 s overlap
- **Software:** EEGLAB toolbox (MATLAB)

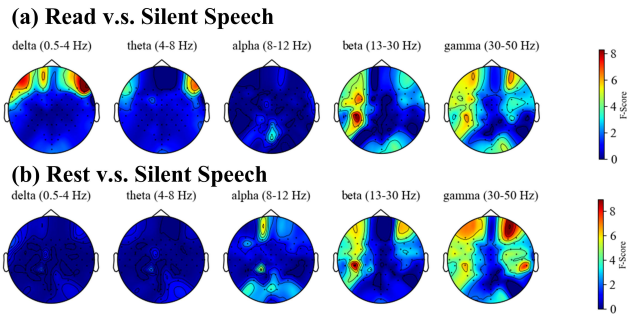


Figure 11: Topographic visualization of F-scores from repeated-measures ANOVA on five canonical EEG frequency bands. (a) Read vs. Silent Speech; (b) Rest vs. Silent Speech. Strong discriminative patterns emerge in the alpha, beta, and gamma bands, particularly in left-lateralized regions.

C.6 Frequency band analysis

To investigate which neural frequency bands are most discriminative for silent speech decoding, we performed a repeated-measures analysis of variance (rm-ANOVA) on the power spectral density (PSD) of EEG signals across five canonical frequency bands: delta (1–4 Hz), theta (4–8 Hz), alpha (8–13 Hz), beta (13–30 Hz), and gamma (30–50 Hz). The analysis compared three condition pairs: rest versus reading, rest versus silent speech, and reading versus silent speech.

F-scores from the ANOVA were used to quantify inter-condition differences, with higher values indicating greater spectral variation. In the delta band, elevated frontal activation was observed, likely reflecting ocular artifacts. The alpha band exhibited moderate but consistent differences in occipital regions, particularly near the POOZ electrode, suggesting changes in visual attention. Notably, the beta and gamma bands showed the most significant activation differences, especially in the left hemisphere. These variations are potentially associated with motor planning and cognitive control mechanisms specific to internal speech generation.

Further comparison between rest and silent speech confirmed that occipital activity diminished, while fronto-temporal beta/gamma activation increased, reinforcing the hypothesis of reduced visual processing and elevated cognitive engagement during silent articulation. These findings highlight alpha, beta, and gamma bands as the most informative for discriminating silent speech states, motivating their

Table 6: Summary of SS-EEG dataset configuration and vocabulary stimuli with syllable counts.

Data Collection Settings	**Stimuli and Semantic Groups**
Subjects: 12	**Motion:** Jumping (2), Running (2), Swimming (2), Going (2)
Modality: EEG	**Emotion:** Happy (2), Sad (1), Fun (1), Horrible (3), College (3)
EEG Channels: 122	**Location:** Home (2), Battlefield (3), Here (2)
Sessions / Subject: 16	**People:** Mother (2), Cowboy (2), Professor (3), Me (1)
Trials / Subject: 6000	**Number:** One (1), Three (1), Eleven (3), Million (2)
Vocabulary Size: 24	**Object:** Spoon (1), Alfa (2), Python (2), Telephone (3)
Total Time / Subject: 10 hrs	

selection in our downstream modeling and multi-band fusion analyses.

Cortical Parcellation

D.1 Anatomical definitions of the 7 RoIs

To incorporate neuroscientific priors into our EEG modeling framework, we partition the scalp-level EEG channels into anatomically informed functional regions of interest (ROIs). These partitions are based on the standard 10–10 international system and follow spatially adjacent topographies consistent with prior literature on cortical function and EEG spatial layout.

We define seven anatomically informed regions of interest (RoIs) based on standard EEG electrode locations. The detailed channel assignments for each region are listed in Table 7.

D.2 Simplified 5-region variant (RoI_5)

For comparison purposes, we also introduce a coarse-grained 5-region mapping by merging Prefrontal and Frontal, and Left/Right Temporal into one Temporal group:

- **Frontal (merged):** Frontal_front + Frontal_middle_and_rear (combined 23 channels)
- **Central, Parietal, Occipital:** Same as RoI_7
- **Temporal:** Combined left and right temporal regions (14 channels)

This version, referred to as BRAINSTACK_ROI5, is used in ablation studies to evaluate the importance of fine-grained spatial decomposition.

Model Architecture and Training Settings

Our model BrainStack integrates both global and region-specific neural representations through a heterogeneous ensemble of expert networks. It consists of: a). 7 region-specific branches built upon CNet, each processing a distinct subset of EEG channels, b). 1 global branch sing a CTNet encoder operating on the union of all EEG channels, and c). gated expert fusion module, which adaptively combines all branch outputs through a learnable attention mechanism and a meta-learner. The input is a segment of EEG signals of shape `(batch_size, 122, 1000)`.

E.1 Global and regional architectures

CTNet processes all available EEG channels and includes:

- Patch Embedding: Conv2D + AvgPool $\rightarrow$ 40-d embeddings.
- Transformer Encoder: 2 layers, 4 heads, dropout=0.5.
- Fully Connected: Flatten $\rightarrow$ 32-d hidden vector.
- Final Classifier: Linear(32 $\rightarrow$ 24).

CNet uses a convolution based network with the following configuration:

E.2 Gated expert fusion

We apply a gated fusion strategy:

- Each branch output (32-d) is projected to a shared 224-d space.
- A learnable weight vector is softmax-normalized with temperature $T = 1.5$.
- DropBranch regularization (drop ratio = 0.25) is used during training.
- Fused features are passed to a meta-learner MLP:

```
MLP: Linear(224 -> 512) -> ReLU -> Linear(512 ->
```

E.3 Training configuration and parameters count

E.4 Dynamic multi-objective loss

We propose a dynamic loss function, `DynamicEnsembleLossProgress`, to jointly optimize fused output, global guidance, regional supervision, and inter-branch consistency. The full objective integrates:

- **Fused prediction loss** $\mathcal{L}_{\text{fused}}$: cross-entropy on the final fused logits.
- **Global teacher loss** $\mathcal{L}_{\text{global}}$: cross-entropy from the global branch (used as soft teacher).
- **Per-region classification loss** $\mathcal{L}_{\text{region}}$: averaged cross-entropy across 7 region-specific branches.
- **Distillation loss** $\mathcal{L}_{\text{distill}}$: KL divergence from global predictions to each region's logits.

Table 7: EEG channel assignments for 7-region cortical division (RoI-7).

RoI	Region Name	#Channel	Assigned EEG Channels
RoI_1	Frontal_front	8	AF3, FP1, FPz, AFz, AF7, AF4, Fp2, AF8
RoI_2	Frontal_middle_and_rear	15	FFT7h, F5, F7, AFF5h, F3, FFC1h, F1, Fz, FFC2h, F2, AFF6h, F4, F6, F8, FFT8h
RoI_3	Central	39	FC5, FFC5h, FFC3h, FC1, FCz, FCC5h, FC3, FCC3h, FCC1h, FCCz, FC2, FFC4h, FFC6h, FC6, FCC2h, FCC4h, FC4, FCC6h, C5, CCP5h, C3, C1, CCP1h, Cz, CCP2h, C2, C4, CCP6h, C6, TTP7h, TPP5h, CP3, CCP3h, CP1, CP2, CCP4h, CP4, TPP8h, TTP8h
RoI_4	Left-Temporal	7	T9, FT9, FTT9H, T7, TP7, FTT7H, FT7
RoI_5	Right-Temporal	7	FTT8H, FT8, T10, FT10, FTT10H, T8, TP8
RoI_6	Parietal	18	P7, CPP5H, CPP3H, CPP1H, CPPZ, P9, P5, P3, PPO1, PPOZ, CPP2H, CPP4H, CPP6H, P8, PPO2, P4, P6, P10
RoI_7	Occipital	28	P11, PO11, PO9, PPO7, PO3, POO7, POO9H, POO11H, I1, OI1, POO3, PO1, POZ, POOZ, OZ, IZ, P12, PO12, PO10, PPO8, PO4, POO8, POO10H, POO12H, I2, OI2, POO4, PO2

Table 8: Configuration of each CNet expert branch.

Module	Configuration
Temporal Conv	1×32 kernel, $f_1 = 16$, padding=same
Spatial Conv	Depthwise, $d = 2$, grouped by f_1
Separable Conv	1×16 kernel, $f_2 = 32$
Pooling	AvgPool (1×8), (1×16), Dropout(0.5)
Classifier	Linear($\cdot \to 24$), max-norm

Table 9: Training configuration and dataset statistics.

Setting	Value
Optimizer	SGD
Scheduler	ReduceLROnPlateau
Learning Rate	5e-3
Batch Size	16
Max Epochs	100
Epochs	100 (early stop: 5)
Weight Decay	1e-4
Validation Split	Random Selection
Num Classes	24
Total Parameters	1.05M
Device	NVIDIA A100 (40GB)

Progress-Based Scheduling. Each term is weighted based on training progress $p \in [0, 1]$, computed as:

$$\text{progress} = \frac{\max(0, \text{epoch} - \text{warmup})}{\text{transition}} \tag{12}$$

With progress increasing linearly after a warm-up period, the component weights evolve as:

$$\lambda_{\text{fused}} = 0.2 + 0.8 \cdot \text{progress} \tag{13}$$

$$\lambda_{\text{global}} = 1 - \lambda_{\text{fused}} \tag{14}$$

$$\alpha = \max_\alpha \cdot \left(1 - \frac{\mathcal{L}_{\text{fused}}}{\mathcal{L}_{\text{max}}}\right) \cdot \text{progress} \tag{15}$$

$$\beta = \max_\beta \cdot \left(1 - \frac{\mathcal{L}_{\text{fused}}}{\mathcal{L}_{\text{max}}}\right) \cdot \text{progress} \tag{16}$$

The final objective is computed as:

$$\mathcal{L}_{\text{total}} = \lambda_{\text{fused}} \cdot \mathcal{L}_{\text{fused}} + \lambda_{\text{global}} \cdot \mathcal{L}_{\text{global}} + \beta \cdot \mathcal{L}_{\text{region}} + \gamma \cdot \mathcal{L}_{\text{distill}} \tag{17}$$

Default Settings. In all experiments, we set:

- `max_alpha = 0.5`, `max_beta = 1.0`
- `warmup_epochs = 5`, `transition_epochs = 5`

This design enables the model to begin with strong global guidance and gradually shift focus toward regional refinement and inter-branch distillation.

Limitations and Future Work

Our results demonstrate that incorporating anatomically grounded brain region parcellation significantly improves EEG-based silent speech decoding. The ensemble modeling of localized neural dynamics provides more robust and interpretable representations compared to global-only approaches. However, several limitations remain.

First, while we adopted a predefined 7-region cortical mapping based on standard EEG montages and prior neuroscience, it is unclear whether this represents the optimal

Table 10: Detailed parameter breakdown of BrainStack components.

Component	# Parameters	Notes
Each CNet (×7)	8.4K–13.3K	Varies by regional input size (total ∼84K)
Global CTNet	∼871K	Includes Patch Embedding, 2-layer Transformer, FC, Output
Feature Projector	7.4K	Linear(32 → 224) projection
Meta-learner MLP	127.5K	Fusion: Linear(224→512→24)
Total	**1.06M**	Trainable parameters in full model

granularity or configuration for language decoding tasks. Future work may explore data-driven parcellation, task-specific functional clustering, or subject-adaptive region selection to better align structural priors with decoding performance.

Second, although our region-guided framework improves generalization, substantial variability still exists across subjects, particularly for those with low signal-to-noise ratios. This suggests that static anatomical partitioning may not fully account for individual neural variability. Learning personalized or dynamic region assignments could further enhance robustness.

Finally, our current approach relies solely on anatomical priors for model structure, but additional domain knowledge—such as temporal dynamics of language processing, hemispheric asymmetry, or neurosemantic pathways—remains underutilized. Integrating such priors through structured inductive biases or neuro-informed architectures is a promising direction for future research.

Ethics

This study was conducted under strict adherence to ethical standards for research involving human participants. All procedures were approved prior to data collection, and measures were taken to ensure participant safety, privacy, and informed participation.

1. Ethics Approval and Participant Consent The experimental protocol was reviewed and approved by the Institutional Human Research Ethics Committee at our institution. All participants were recruited voluntarily and provided written informed consent. Prior to participation, subjects were fully briefed on the nature of the silent speech EEG experiment, including the tasks, session duration, and the intended research goals. Participants were explicitly informed of their right to withdraw at any time without consequence.

2. Data Privacy and Anonymization No personally identifiable information (PII) was collected. All EEG data were anonymized upon recording and stored on encrypted, access-controlled servers. Data usage was strictly limited to academic research purposes. The anonymization process ensures that individual identities cannot be recovered, protecting the privacy of participants throughout all stages of data handling and analysis.

3. Physical Safety and Intended Use The experimental procedure involved only non-invasive EEG measurements and silent articulation tasks, posing minimal physical or psychological risk. Participants were given sufficient breaks to avoid fatigue. This work is intended to advance assistive communication and neurotechnology research. The dataset and models are not designed for inferring internal cognitive states without consent. Any future application of this technology must adhere to established ethical norms regarding transparency, privacy, and non-coercive use.

Broader Impacts

H.1 Positive Impacts

Assistive Communication for Speech-Impaired Users Our EEG-based silent speech decoding framework offers a non-invasive pathway for individuals with severe speech impairments to express themselves. This technology can serve as a foundation for future communication aids, enabling users to silently articulate intended words and improving quality of life, autonomy, and social participation.

Neurotechnological Advancement By linking localized cortical activity to linguistic representations, our work contributes to the broader understanding of neural language processing. It provides empirical evidence for the role of anatomically defined brain regions in internal speech decoding, which may inform clinical diagnostics and cognitive neuroscience research.

Scalable and Accessible BCI Design The BrainStack architecture is lightweight and compatible with widely available EEG hardware, reducing barriers to real-world deployment. Its modular structure also allows future integration with adaptive or multilingual systems, expanding the reach of BCI research to more diverse user groups and application domains.

H.2 Negative Impacts

Privacy, Consent, and Misuse Risk Decoding neural signals—especially internal speech—raises significant concerns around user privacy, informed consent, and the potential for surveillance. Improper collection, storage, or interpretation of EEG data could unintentionally reveal sensitive cognitive states or user identities. Moreover, BCI systems developed for assistive purposes could be repurposed for covert behavioral inference or unauthorized monitoring. While our current system is limited to word-level decoding under controlled conditions, future models with higher resolution and broader vocabulary coverage may amplify these

risks. Ethical deployment requires robust data governance, transparency, and strict usage constraints to prevent misuse.

Bias and Representational Limitations Our dataset is derived from 12 healthy, right-handed English speakers, limiting its demographic and linguistic diversity. This raises concerns about potential performance disparities when applied to broader populations, including users with speech or neurological impairments, non-native speakers, or those with atypical cortical patterns. Without inclusive data collection and fairness-aware training objectives, BCI systems risk encoding existing biases or failing to generalize across user groups. Future work should prioritize equitable representation, demographic balancing, and bias auditing in both data and modeling.

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